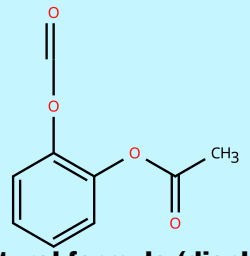
ASPIRIN

Common medicine Potent efficacy

Compound Identity

Common name : Aspirin
IUPAC name : ACETYLSALICYLIC ACID
Molecular formula : $C_9H_8O_4$
molecular weight : 180.16 g/mol

Structure



Structural formula (displayed)

Functional Groups

- Carboxylic acid (-COOH)
- Ester (-COOR'),
- Aromatic ring (benzene derivative)

Notable feature

• Carboxylic acid and ester attached to a benzene ring.

Important Properties

Appearance / State
White, crystalline solid.

Solubility
Poorly soluble in water; soluble in ethanol.

Boiling Point
135 °C

Melting Point
Decomposes above melting point.

pKa
~3.5 (Weak acid)

What is Amazing About It?

Aspirin is one of the most widely used analgesics and anti-inflammatory drugs in the world!

Its main medicinal effect is to reduce pain and inflammation by inhibiting enzymes, and it is crucial for preventing heart attacks.

Uses / Value

Everyday uses
- Analgesic (pain relief);
- Antipyretic (fever reducer);
- Anti-inflammatory

Medical uses
- Reduces risk of heart attacks and strokes;
- Antiplatelet effect (prevents blood clots)

Industrial uses
- Studied for cancer risk reduction.

Chemical engineering connec
Produced via the acetylation of salicylic acid. Process design optimizes high-purity yield.

Risks & Safety

Health risks
- Gastrointestinal bleeding and stomach irritation;
- Reye's syndrome in children with viral infections.

Allergies
- Aspirin intolerance or sensitivity.

Fire / Reactivity
- Tinnitus (ringing in the ears), dizziness, metabolic issues.

Environmental concerns
- Detected in wastewater; proper disposal is important.

References

1. National Center for Biotechnology Information. (n.d.). Aspirin. PubChem. https://pubchem.ncbi.nlm.nih.gov/compound/Aspirin
2. National Library of Medicine. (n.d.). PubChem. https://www.nlm.nih.gov/oet/ed/navigator/pubchem1/index.html

A medical marvel, used for centuries.

